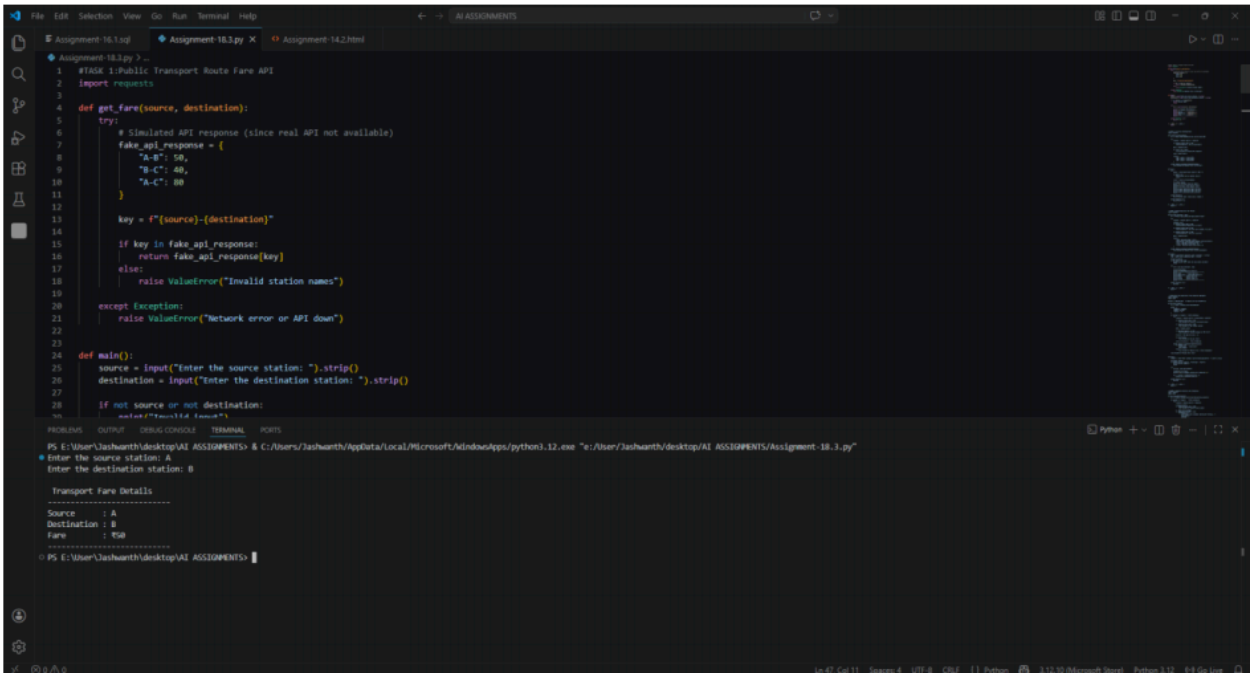
ASSIGNMENT-18.3
ROLL NO: 2303A51062
BATCH: 29

TASK-1:
PROMPT:
Generate Python code using requests library to fetch fare between two stations using a
public transport API. Validate inputs, handle invalid stations, and API failures. Display
result clearly.
CODE:

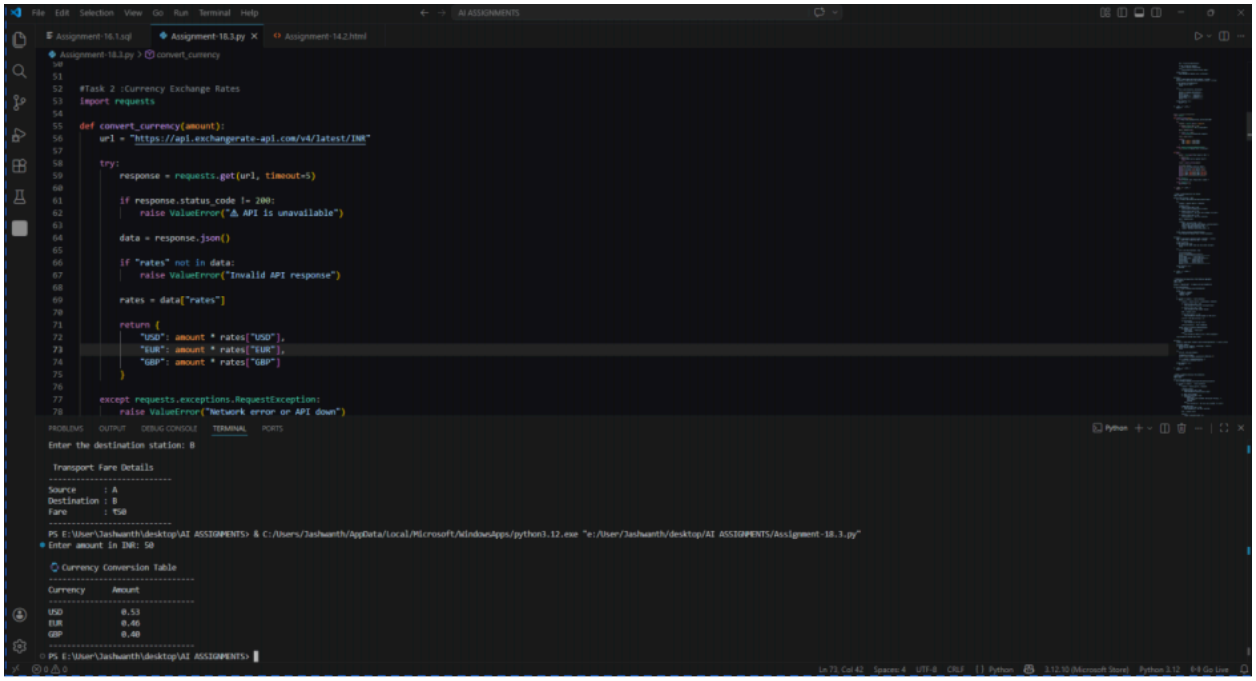


```
#TASK 1:Public Transport Route Fare API
import requests

def get_fare(source, destination):
    try:
        # Simulated API response (since real API not available)
        fake_api_response = {
            "A-B": 50,
            "B-C": 40,
            "A-C": 80
        }

        key = f"{source}-{destination}"

        if key in fake_api_response:
            return fake_api_response[key]
        else:
            raise ValueError("Invalid station names")

    except Exception:
        raise ValueError("Network error or API down")


def main():
    source = input("Enter the source station: ").strip()
    destination = input("Enter the destination station: ").strip()

    if not source or not destination:
```

```
PS E:\User\lashwanth\desktop\AI ASSIGNMENTS> & C:/Users/lashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/lashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter the source station: A
Enter the destination station: B

 Transport Fare Details
-----------------------
Source      : A
Destination : B
Fare        : ₹50
-----------------------
PS E:\User\lashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATION:
The program checks that the station names entered are valid before making the request.
If the API fails or the stations don’t exist, it shows a clear error message. When
everything works, the fare is displayed in a simple and easy-to-read format.

TASK-2:
PROMPT:
Generate Python function to fetch GitHub repository details using GitHub API. Handle
404 errors, rate limits, and invalid input. Display repo details clearly
CODE:



```
#Task 2 :Currency Exchange Rates
import requests

def convert_currency(amount):
    url = "https://api.exchangerate-api.com/v4/latest/INR"

    try:
        response = requests.get(url, timeout=5)

        if response.status_code != 200:
            raise ValueError("⚠ API is unavailable")

        data = response.json()

        if "rates" not in data:
            raise ValueError("Invalid API response")

        rates = data["rates"]

        return {
            "USD": amount * rates["USD"],
            "EUR": amount * rates["EUR"],
            "GBP": amount * rates["GBP"]
        }

    except requests.exceptions.RequestException:
        raise ValueError("Network error or API down")
```

```
Enter the destination station: B

 Transport Fare Details
--------------------------
Source      : A
Destination : B
Fare        : ₹50
--------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter amount in INR: 50

 Currency Conversion Table
--------------------------------
Currency      Amount
--------------------------------
USD           0.53
EUR           0.46
GBP           0.40
--------------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATION:
The function connects to GitHub and retrieves details like the repository name,
description, stars, forks, and issues. If the repository doesn’t exist or the API limit is
reached, it shows a helpful error message. The output is printed in a neat and structured
Way.

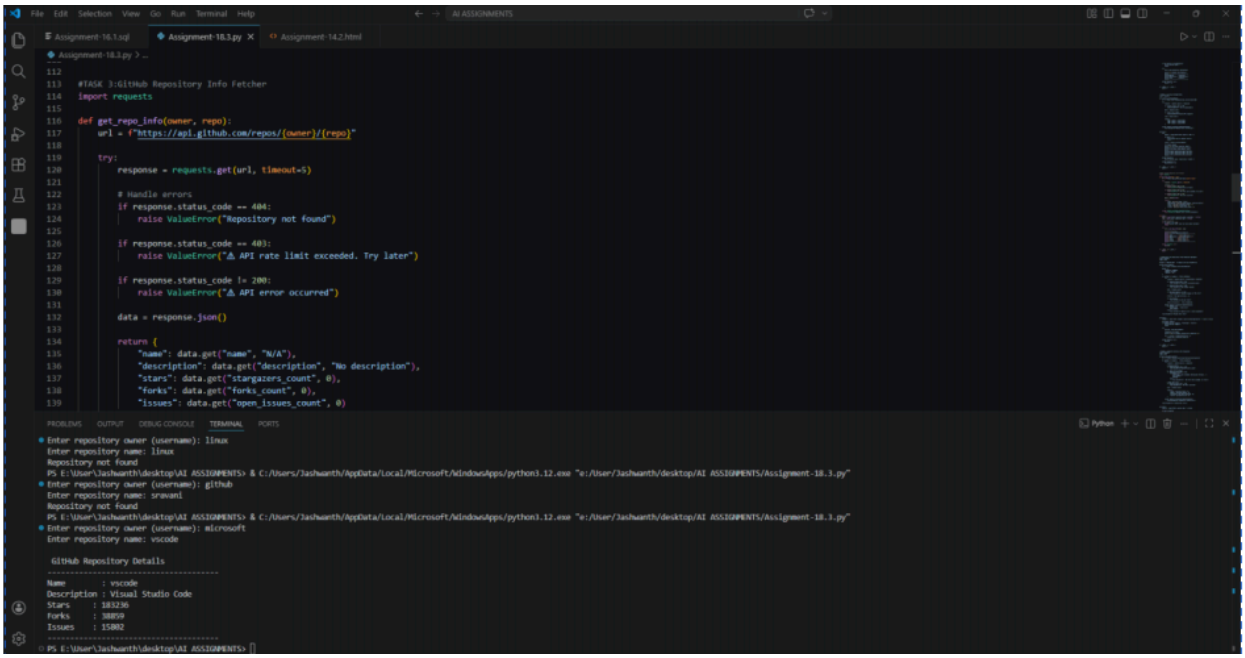
TASK-3:

PROMPT:

Write Python script using News API to fetch top 5 headlines by category. Handle invalid

category, API key issues, and request failures. Add retry logic.

CODE:



```
#TASK 3:GitHub Repository Info Fetcher
import requests

def get_repo_info(owner, repo):
    url = f"https://api.github.com/repos/{owner}/{repo}"

    try:
        response = requests.get(url, timeout=5)

        # Handle errors
        if response.status_code == 404:
            raise ValueError("Repository not found")

        if response.status_code == 403:
            raise ValueError("⚠ API rate limit exceeded. Try later")

        if response.status_code != 200:
            raise ValueError("⚠ API error occurred")

        data = response.json()

        return {
            "name": data.get("name", "N/A"),
            "description": data.get("description", "No description"),
            "stars": data.get("stargazers_count", 0),
            "forks": data.get("forks_count", 0),
            "issues": data.get("open_issues_count", 0)
```

```
Enter repository owner (username): linux
Enter repository name: linux
Repository not found
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter repository owner (username): github
Enter repository name: sravani
Repository not found
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter repository owner (username): microsoft
Enter repository name: vscode

 GitHub Repository Details
----------------------------------------
Name        : vscode
Description : Visual Studio Code
Stars       : 183236
Forks       : 38859
Issues      : 15802
----------------------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATION:
The script fetches the latest headlines based on the chosen category. If the category is
wrong or the API key is invalid, it alerts the user. It also retries the request if there's a
temporary issue, ensuring that the user still gets the news.

TASK-4:

PROMPT:

Write Python script using News API to fetch top 5 headlines by category. Handle invalid
category, API key issues, and request failures. Add retry logic.

CODE:

```python
#TASK4:Real-Time Application: News Headlines Aggregator
import requests
import time

API_KEY = "YOUR_API_KEY"   # Replace with your NewsAPI key

def get_news(category):
    url = "https://newsapi.org/v2/top-headlines"

    params = {
        "category": category,
        "apiKey": API_KEY,
        "country": "in"
    }

    for attempt in range(2):  # Retry mechanism
        try:
            response = requests.get(url, params=params, timeout=5)

            if response.status_code == 401:
                raise ValueError("Invalid or missing API key")

            if response.status_code != 200:
                raise ValueError("API request failed")

            data = response.json()
```

```
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter repository owner (username): github
Enter repository name: sravani
Repository not found
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter repository owner (username): microsoft
Enter repository name: vscode

 GitHub Repository Details
----------------------------------
Name        : vscode
Description : Visual Studio Code
Stars       : 183236
Forks       : 38859
Issues      : 15802
----------------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter category (sports/technology/health): technology
Invalid or missing API key
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

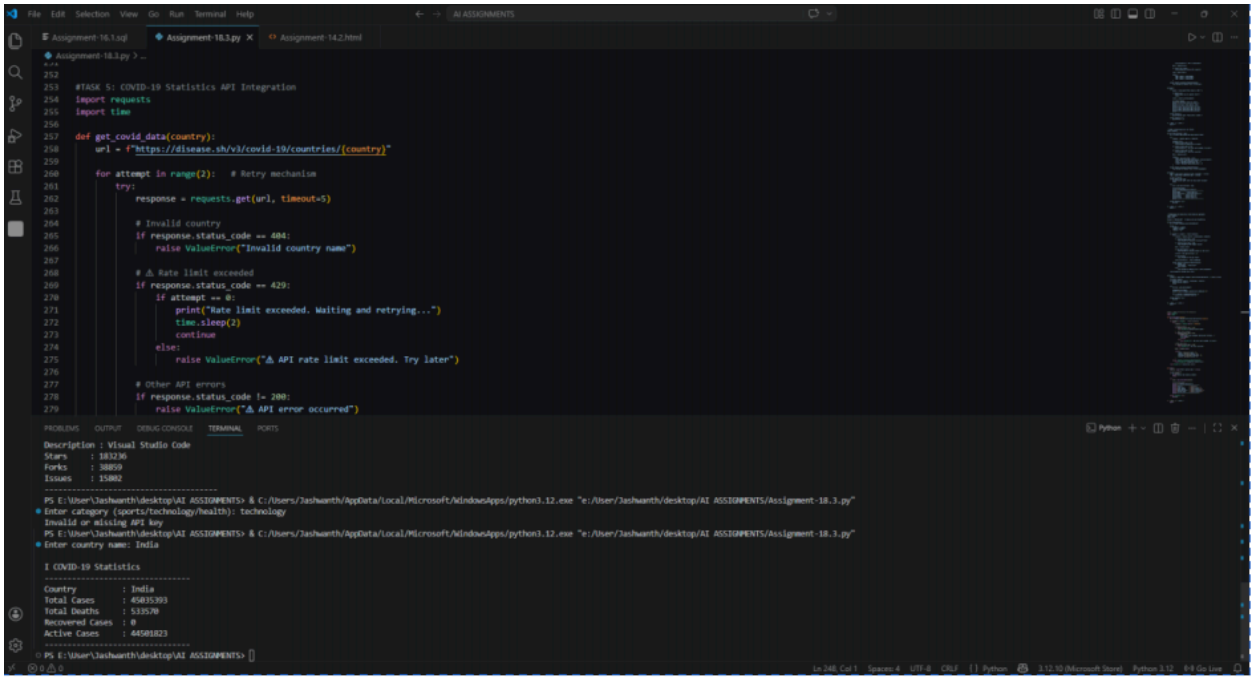
OBSERVATION:
The program displays the top 5 news headlines clearly. It manages errors like invalid
categories or API problems gracefully, and the retry logic helps recover from temporary
failures. This makes the script reliable for fetching news updates.

TASK-5:

PROMPT:

Generate Python code to fetch COVID-19 stats by country using requests. Handle
invalid country, rate limits, and network errors. Include retry logic.

CODE:



```python
#TASK 5: COVID-19 Statistics API Integration
import requests
import time

def get_covid_data(country):
    url = f"https://disease.sh/v3/covid-19/countries/{country}"

    for attempt in range(2):   # Retry mechanism
        try:
            response = requests.get(url, timeout=5)

            # Invalid country
            if response.status_code == 404:
                raise ValueError("Invalid country name")

            # ⚠ Rate limit exceeded
            if response.status_code == 429:
                if attempt == 0:
                    print("Rate limit exceeded. Waiting and retrying...")
                    time.sleep(2)
                    continue
                else:
                    raise ValueError("⚠ API rate limit exceeded. Try later")

            # Other API errors
            if response.status_code != 200:
                raise ValueError("⚠ API error occurred")
```

```
Description : Visual Studio Code
Stars       : 183236
Forks       : 38859
Issues      : 35802
-----------------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter category (sports/technology/health): technology
Invalid or missing API key
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-18.3.py"
Enter country name: India

I COVID-19 Statistics
-----------------------------
Country         : India
Total Cases     : 45035393
Total Deaths    : 533570
Recovered Cases : 0
Active Cases    : 44501823
-----------------------------
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATION:

The script retrieves COVID-19 statistics for a given country and shows them in a clear
format. If the country name is wrong or the API fails, it provides an error message. The
retry logic ensures that temporary network issues don’t stop the program from working.